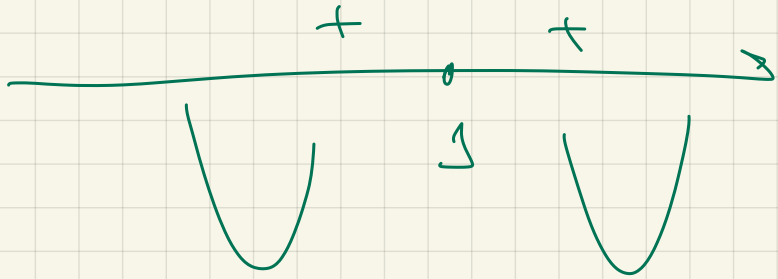
1+

2

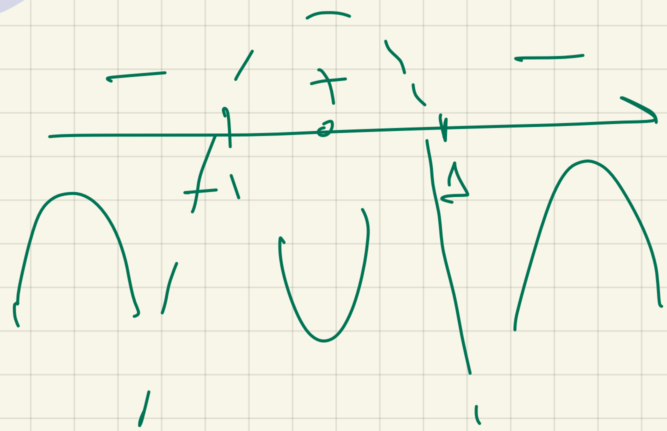


$$f''(x) = \frac{2(1-x^2)}{(1+x^2)^2}$$

$1-x^2=0$

$x^2=1$

$x=\pm 1$



$D(f(x))$

$\cancel{x \neq 0}$

$1+x^2>0$

$x^2>-1$

x любое

$\ln(1-2x) = f(x)$

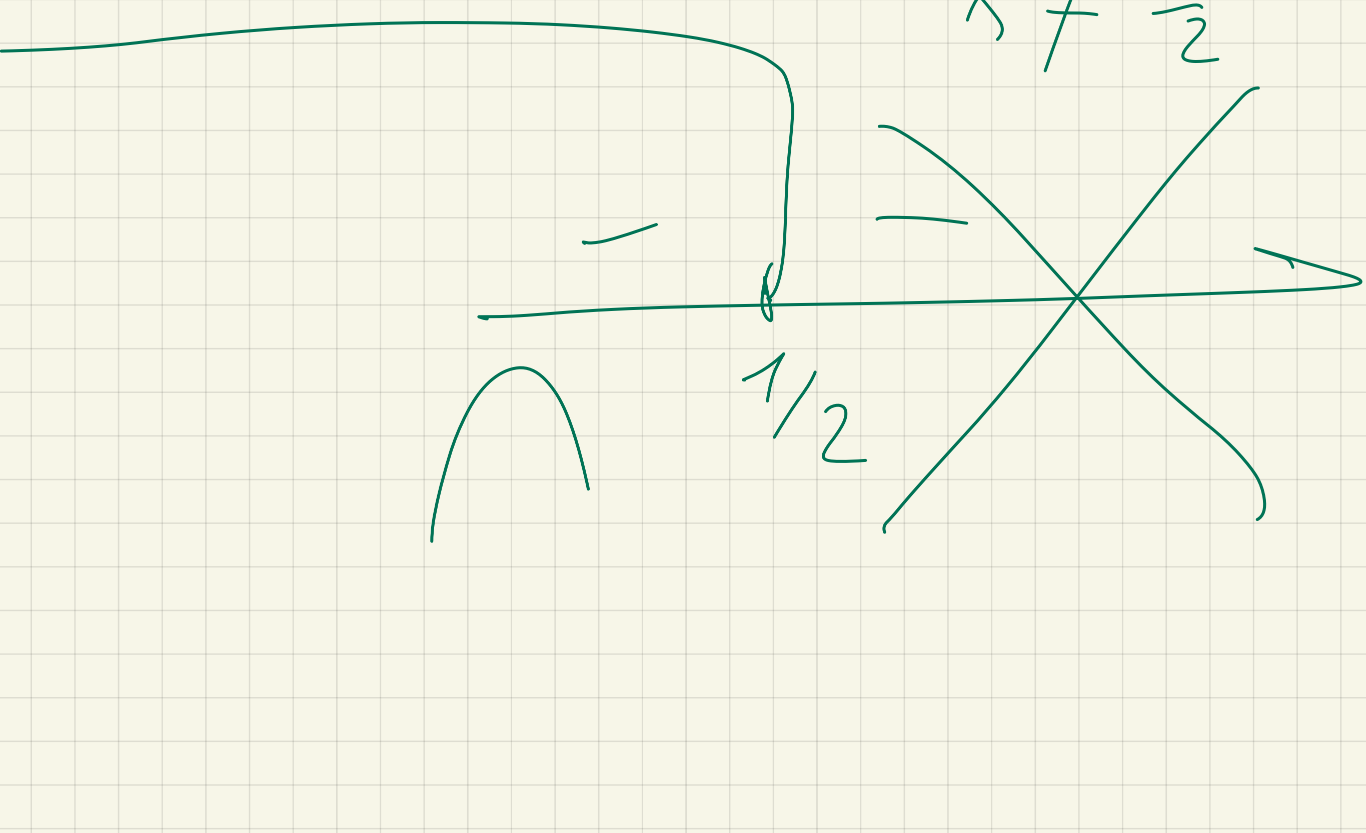
$D/f(x) = 1-2x > 0 \Rightarrow 2x < 1$

$x < \frac{1}{2}$

$f'(x) = \frac{1}{1-2x} \cdot (-2) = \frac{2}{2x-1}$

$f''(x) = \frac{-2 \cdot 2}{(2x-1)^2} = \frac{-4}{(2x-1)^2}$

$x \neq \frac{1}{2}$



$$\ln(1-4x^2) = f(x)$$

$$1-4x^2 > 0$$

$$4x^2 < 1$$

$$x^2 < \frac{1}{4}$$

$$-\frac{1}{2} < x < \frac{1}{2}$$

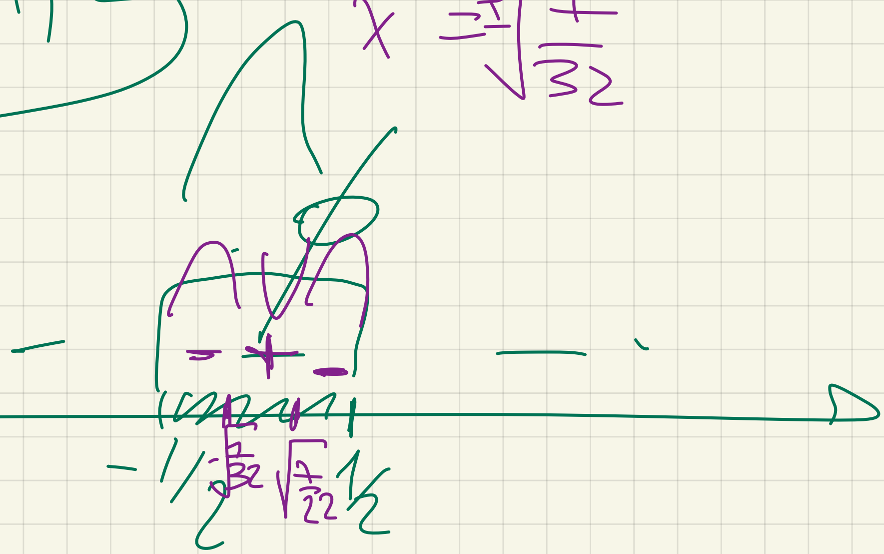
$$f'(x) = \frac{1}{1-4x^2} \cdot (-8x) = \frac{8x}{4x^2-1}$$

$$f''(x) = \frac{8(4x^2-1) - 8x(8x)}{(4x^2-1)^2}$$

$$\frac{-32x^2 + \cancel{8}7}{(4x^2-1)^2} = 0$$

$$x = \pm\sqrt{\frac{7}{32}}$$

$$-32x^2 = 8$$



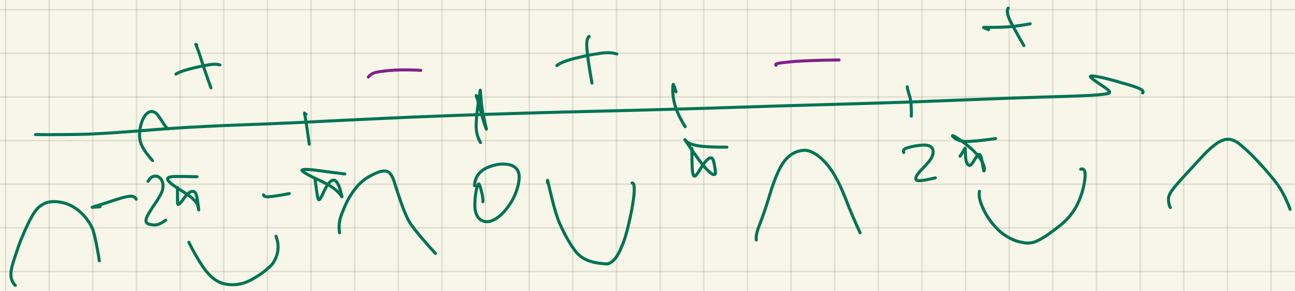
$f(x) = x - \sin x$

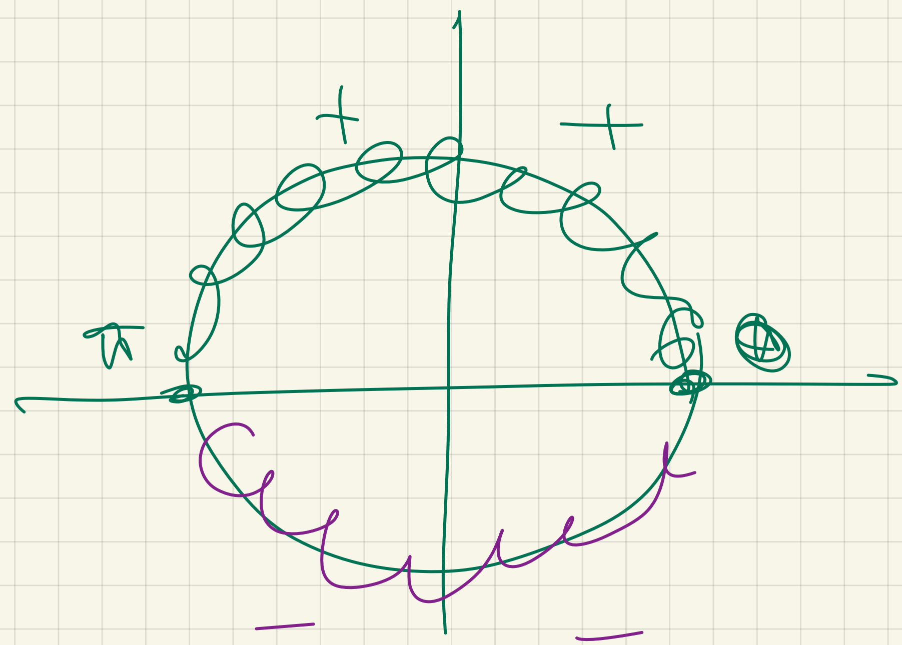
$f'(x) = 1 - \cos x$

$f''(x) = \sin x = 0$

$x = 0$

$x = \pi k \quad k \in \mathbb{Z}$





$$f(x) = \frac{x}{1+x^2}$$

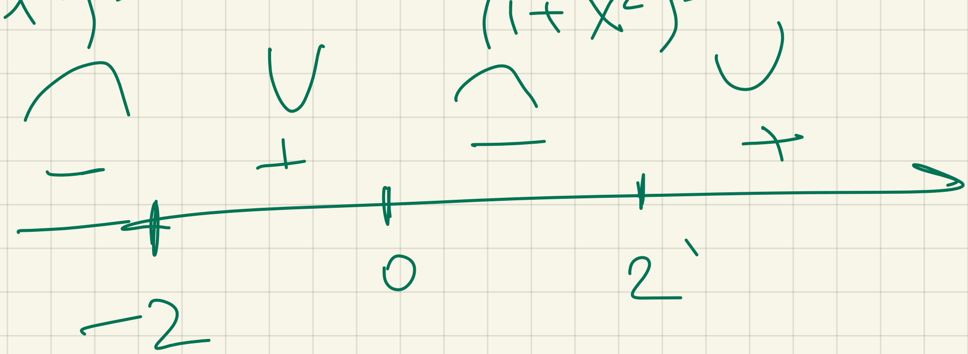
$$f'(x) = \frac{1(1+x^2) - 2x \cdot x}{(1+x^2)^2} = \frac{-x^2+1}{(1+x^2)^2}$$

$$f''(x) = \frac{-2x(1+x^2)^2 - 2(1+x^2) \cdot 2x \cdot (1-x^2)}{((1+x^2)^2)^2}$$

$$\frac{(1+x^2)(-2x(1+x^2 - 2 \cdot 2x \cdot (1-x^2)))}{(1+x^2)^4}$$

$$= \frac{-2x - 2x^3 - 4x + 4x^3}{(1+x^2)^3} =$$

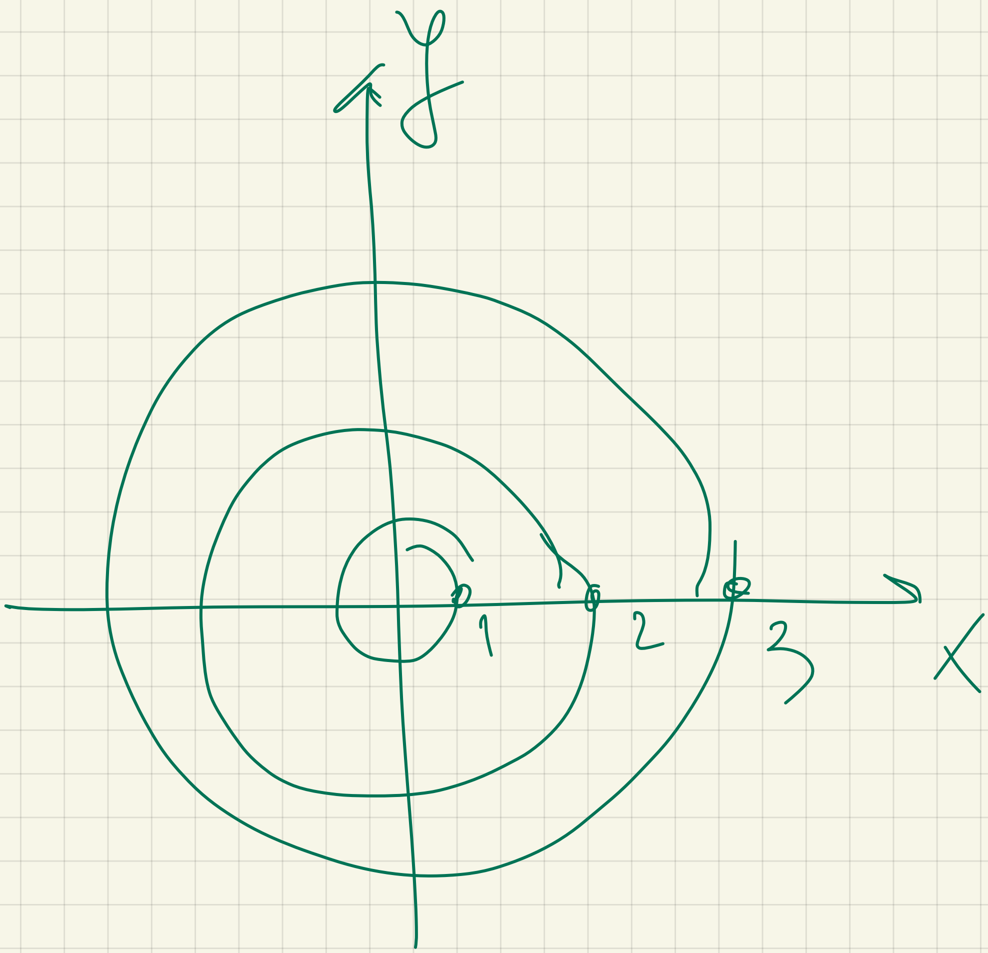
$$= \frac{2x^3 - 8x}{(1+x^2)^3} = \frac{2x(x^2-4)}{(1+x^2)^3}$$

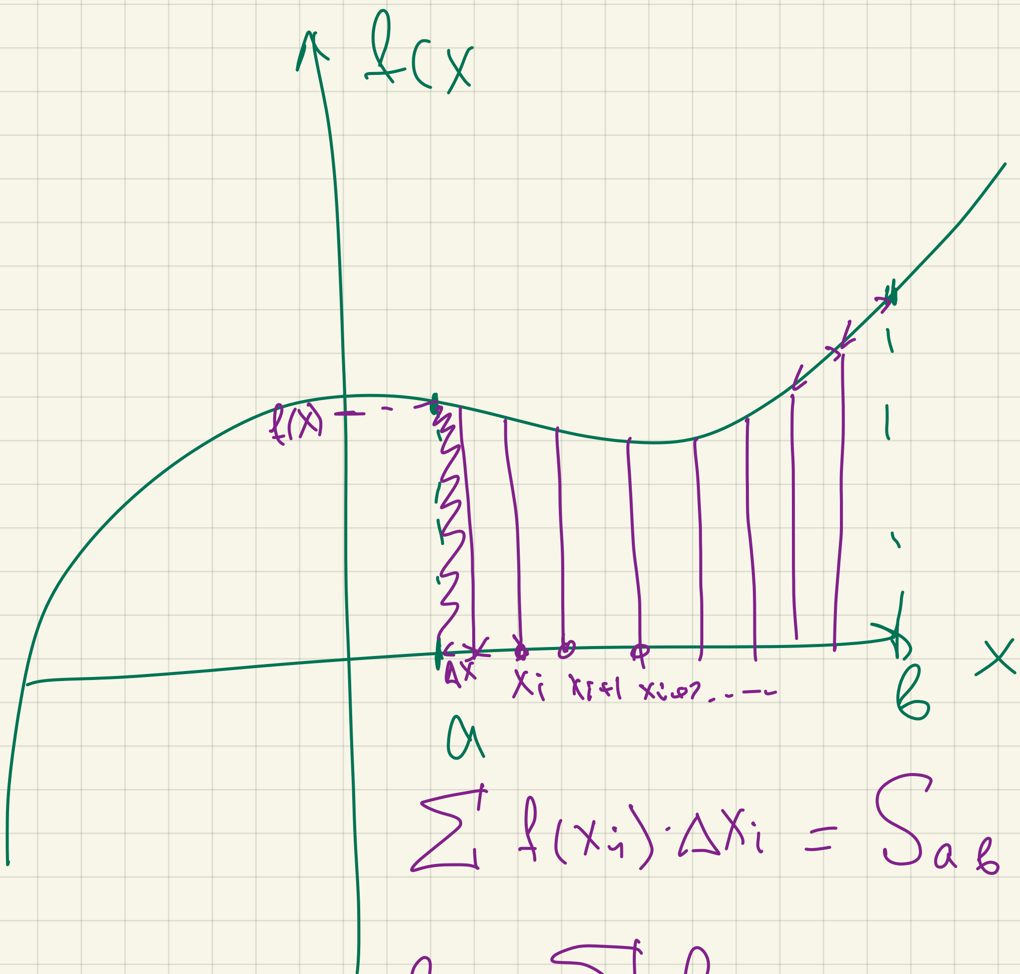


$$e^{-(x^2+y^2)}$$

$$e^{-1} \quad e^{-4} \quad e^{-9}$$

$$\begin{cases} -(x^2+y^2) = -1 \\ -(x^2+y^2) = -4 \\ -(x^2+y^2) = -9 \end{cases} \Rightarrow \begin{array}{l} x^2+y^2=1 \\ x^2+y^2=4 \\ x^2+y^2=9 \end{array}$$





$$\sum f(x_i)\cdot\Delta x_i = S_{ab}$$

$$\lim_{\Delta x\to 0}\sum f(x)\cdot\Delta x = \int_a^b f(x)\,dx = S_{ab}$$

$x^2 - 2x + 5$

$x^2 - 2x - 1000$

$$f(x) = x^2 - 2x + \underline{\underline{1}}$$

$$f'(x) = 2x - 2 =$$

$$\int f'(x)\,dx = f(x) + C$$

$f(b) - f(a)$

множество констант

Ньютона - Лейбница

$$\int_a^b f(x)\,dx = \underline{\underline{F}}(b) - \underline{\underline{F}}(a)$$

первообр.

$f(x) = x^2 - 2x + 1$ $\quad a = 1$

$f'(x) = 2x - 2$ $\quad b = 2$

$$\int_1^2 2x - 2 = x^2 - 2x + C \Big|_1^2 =$$

$4 - 4 + c - (1 - 2 + c) =$

$4 - 4 + \cancel{c} - 1 + 2 - \cancel{c} = 1$

$$\begin{cases} \int f'(x)\,dx = f(x) + C \\ f(1) = A = 10 \end{cases}$$

$f(x) = x^2 - 2x + C$

$f(1) = 10$

$\cancel{1}^2 - 2\cancel{1} + C = 10$

$C = 10 - 1 + 2 = 11$

$f(x) = x^n \longrightarrow F(x) = \frac{x^{n+1}}{n+1}$

$f(x) = k \qquad F(x) = kx$

$f(x) = \frac{1}{x} \qquad F(x) = \ln x$

$f(x) = e^x \qquad F(x) = e^x$

$f(x) = \sin x \qquad F(x) = -\cos x$

$f(x) = \cos x \qquad F(x) = \sin x$

$$\int_1^2 (x^2 + 5x - 3)\,dx = \left. \frac{x^3}{3} + \frac{5x^2}{2} - 3x + C \right|_1^2$$

$$\int (x^3 + 4x)\,dx$$

$f(1) = 0$

$\underline{\underline{F(x)}}$?

$$= \frac{x^4}{4} + \frac{4x^2}{2} + C$$

$$C = -\frac{9}{4}$$

$$F(x) = \frac{x^4}{4} + \frac{4x^2}{2} - \frac{9}{4}$$

$$x^3 + 4x$$

$$MC(Q) = \frac{2}{Q}\left(8Q^2 - 3Q^3 + Q^4\right) + 1$$

$$C(1) = 17$$

$$\underline{\underline{C(Q)}} = \int MC(Q) =$$

$$\int \left(\frac{2\left(8Q^2 - 3Q^3 + Q^4\right)}{Q} + 1 \right) dQ =$$

$$C(Q)$$

$$\int (16Q - 3Q^2 + 2Q^3 + 1)\, dQ \Rightarrow$$

$$\Rightarrow C(Q) = \frac{16Q^2}{2} - \frac{3Q^3}{3} + \frac{2Q^4}{4} + Q + C$$

$$= 8Q^2 - Q^3 + \frac{Q^4}{2} + Q + C$$

$$C(1) = 17$$

$$8 - \cancel{1} + \frac{1}{2} + \cancel{1} + C = 17$$

$$\frac{17}{2} + C = 17$$

$$C = 17 - \frac{17}{2} = \frac{17}{2}$$

$$C(Q) = 8Q^2 - Q^3 + \frac{Q^4}{2} + Q + \frac{17}{2}$$

$AC(Q)$

$Q = 2$

$MC(Q) = Q(6Q^2 - 2Q^3 + 3Q^4) + 2$

$C(0) = 5$

$C(Q)$?

$$C(Q) = \frac{6Q^4}{4} - \frac{2Q^5}{5} + \frac{3Q^6}{6} + 2Q + C$$

$$C(0) = C = 5$$

$$\frac{3Q^4}{2} - \frac{2Q^5}{5} + \frac{Q^6}{2} + 2Q + 5 = C(Q)$$

$$\lim_{x \to 0} \frac{e^{2026x} - 1}{\ln(2025x + 1)} = \lim_{x \to 0} \frac{e^{2026x} \cdot 2026}{\frac{2025}{2025x+1}}$$

$$\lim_{x\to 0} \frac{2026}{2025} \underbrace{e^{2026x}}_{e^0 = 1} \underbrace{\left(1+2025x\right)}_{\frac{1+0}{1}} =$$

$$\frac{2026}{2025}$$